Sparse data limit what a corrosion model can predict

Identifiability of a point defect model for oxide growth on AISI 347

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The problem

Mechanistic corrosion models are routinely

- written with four to six adjustable parameters,
- fitted to a handful of measurements,
- and then extrapolated across a service lifetime.

What is almost never tested

Whether the data determine those parameters at all.

A model that fits is not the same as a model that is *identified*. If several very different parameter sets fit equally well, the fit is reported honestly and the extrapolation is still meaningless.

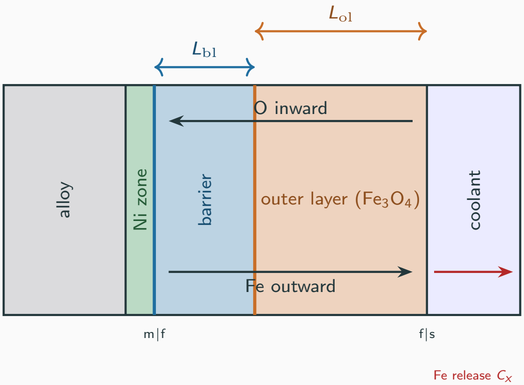
The system

Nb-stabilised austenitic stainless steel **X6CrNiNb18-10** (AISI 347)

Simulated boiling-water-reactor hydrothermal water

- 240 °C, 7 MPa
- 0.4 ppm dissolved O_2
- exposures 72, 168, 480 h

A **duplex** oxide forms: a thin Cr-rich barrier layer that controls the kinetics, and a thicker, far more variable Fe-rich outer layer.



The barrier layer sets the kinetics; the outer layer is where the information runs out.

All of the calibration data

t (h)	Cr barrier (nm)	Fe outer (nm)	scans
72	37.3 ± 7.7	143.3 ± 111.5	3
168	98.3 ± 21.7	225.5 ± 142.7	4
480	134.3 ± 5.4	240.3 ± 172.5	3

Six numbers

Three Cr, three Fe. Five fitted parameters.

Extrapolated 180-fold

Last observation 480 h $\rightarrow$ target 87 600 h.

The Fe standard deviations are 50 to 70% of the means. Whatever the model says about the outer layer, the data barely constrain it.

The reduced two-layer model

Barrier layer, from point defect kinetics with a field that relaxes as the film thickens:

$$\frac{dL_{\mathrm{bl}}}{dt} = A_{\mathrm{bl}}\, e^{\,b_3 L_{\mathrm{bl}}} - C_{\mathrm{bl}}, \qquad L_{\mathrm{bl}}(0) = L_0.$$

Outer layer, fed by cations crossing the barrier and losing mass to the coolant:

$$\frac{dL_{\mathrm{ol}}}{dt} = \mathrm{PBR}_{\mathrm{eff}}\,\frac{dL_{\mathrm{bl}}}{dt} + C_x, \qquad L_{\mathrm{ol}}(0) = L_{\mathrm{ol},0}.$$

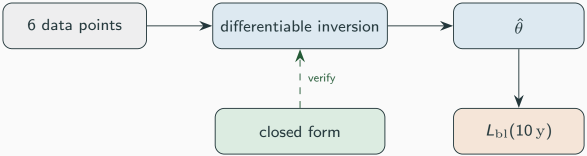
One-way coupling, and why it matters

L_{bl} never reads $\mathrm{PBR}_{\mathrm{eff}}$, C_x or $L_{\mathrm{ol},0}$. Every barrier-layer result, including the ten-year prediction, is structurally immune to everything the outer layer does. This is what lets us report an outer-layer non-result without contaminating the headline number.

Closed form, and what we verify against it

With $C_{\mathrm{bl}} = 0$ the barrier equation integrates exactly:

$$L_{\mathrm{bl}}(t) \;=\; \frac{1}{b_3}\,\ln\!\Big(e^{\,b_3 L_0} + A_{\mathrm{bl}}\,b_3\,t\Big).$$



The numerical solver is validated against the closed form to better than 1×10^{-10} nm, so any misfit we report is a statement about the data and the model, never about the solver.

The question we actually ask

Hypothesis

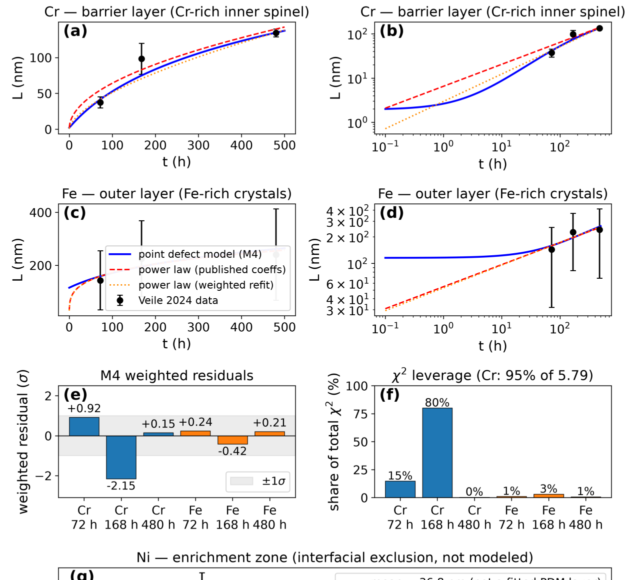
The six measurements are enough to *fit* the reduced point defect model, but not enough to *identify* it; and the resulting extrapolation is dominated by the choice of model form rather than by the data.

Three tests, each capable of falsifying it:

1. **Profile likelihood** across each parameter, with the others re-optimised.
2. **Parametric bootstrap**, 1000 members, resampling at the measured scatter.
3. **Prior relaxation**, to check that a conclusion is not an artefact of a prior.

And one comparison: the same data fitted with a purely empirical power law, then both extrapolated.

Result 1: the model fits



Accepted model **M4**:

$$A_{\mathrm{bl}} = 0.7269\ \mathrm{nm/h}$$
$$b_3 = -0.012487\ \mathrm{nm}^{-1}$$
$$\mathrm{PBR}_{\mathrm{eff}} = 1.096$$
$$L_0 = 1.924\ \mathrm{nm}$$
$$L_{\mathrm{ol},0} = 115.6\ \mathrm{nm}$$

$$\chi^2 = 5.79, \quad \mathrm{dof} = 1$$

The empirical power law gives $\chi^2 = 20.10$ on 2 dof, so the mechanistic model is the better description of *these* points.

Result 2: but the fit is one measurement

Weighted residual decomposition at M4:

	72 h	168 h	480 h
Cr (barrier)	+0.924	**−2.154**	+0.149
Fe (outer)	+0.242	−0.419	+0.210

95%

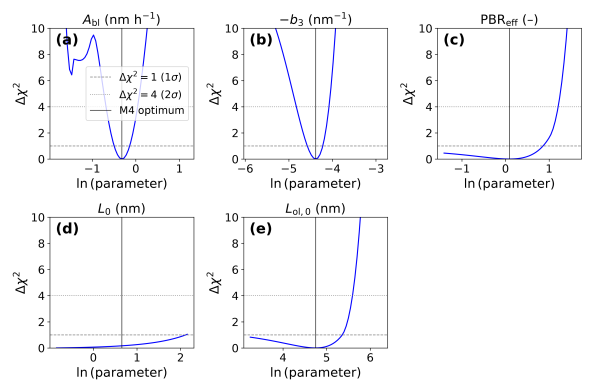
of χ^2 sits on the three chromium points.

80%

sits on the single 168 h chromium point.

A goodness-of-fit statistic reported as a single number hides this completely. The model is being judged, in effect, by one measurement.

Result 3: one parameter is not determined at all



param	max $\Delta\chi^2$	verdict
b_3	129.6	identifiable
$L_{\mathrm{ol},0}$	58.2	identifiable
A_{bl}	50.0	identifiable
$\mathrm{PBR_{eff}}$	19.5	weakly id.
L_0	**1.0**	unidentifiable

L_0's profile is **flat**: the data cannot move it. Its apparent bootstrap precision (1.92 ± 0.04 nm) is the prior speaking, not the measurements.

$\mathrm{PBR_{eff}}$ spans [0.24, 2.32] at 1σ.

Result 4: the outer layer is exactly determined, so it says nothing

3 free parameters

$\mathrm{PBR}_{\mathrm{eff}}$ $\quad C_x \quad L_{\mathrm{ol},0}$

$\updownarrow$ 0 dof

3 observations

Fe at 72, 168, 480 h

exactly determined

$\Rightarrow$ *any* admissible $\mathrm{PBR}_{\mathrm{eff}}$ is reachable

$\mathrm{PBR}_{\mathrm{eff}}$	$\Delta\chi^2$	C_x (nm/h)
1.096 (fit, $C_x = 0$)	—	0
2.053 (mass balance)	−0.120	−0.241
2.156 (with Ni routing)	−0.131	−0.264
2.433 (no precipitate)	−0.157	−0.327

The apparent factor-1.9 discrepancy between the fitted $\mathrm{PBR}_{\mathrm{eff}}$ and the stoichiometric mass balance is **not a discrepancy**. The stoichiometric value costs nothing; χ^2 even improves.

The data cannot separate iron *production* from iron *loss to the coolant*.

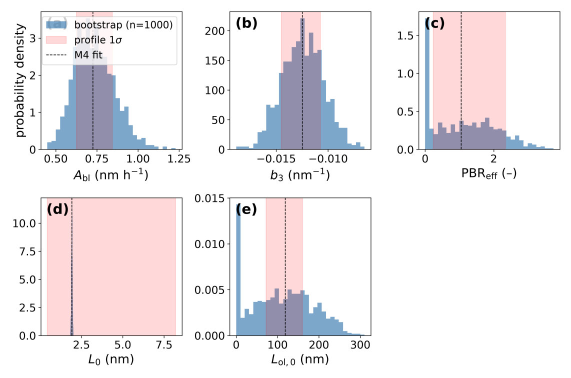
A falsifiable by-product

Imposing the stoichiometric ratio forces a specific outer-layer loss rate, and therefore a specific iron release into the coolant:

$\mathrm{PBR}_{\mathrm{eff}}$ imposed	C_x (nm/h)	Fe release (μg dm^{-2} h^{-1})
1.096	−0.025	0.94
2.156	−0.264	**9.94**
2.433	−0.327	12.29

A coolant-chemistry measurement discriminates these by an order of magnitude. The non-result therefore comes with an experiment that would convert it into a result.

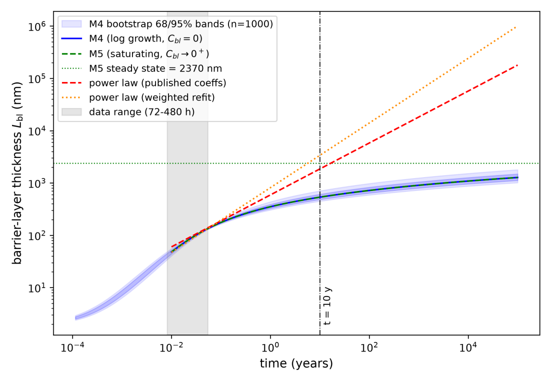
Result 5: uncertainty, propagated honestly

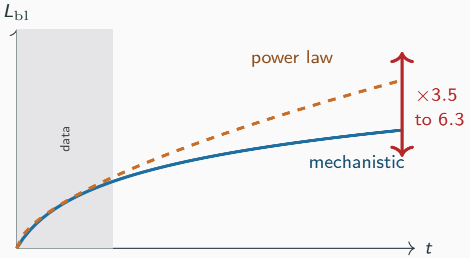


1000 bootstrap members, zero failures. At ten years: $L_{\mathrm{bl}} = \mathbf{535}$ nm, 95% interval [447, 704] nm.

This interval is *conditional on the model form being right*.

Result 6: the key result





Same data, divergent futures

Ten-year barrier thickness

mechanistic	535 nm
power law	1853 to 3375 nm

a factor of **3.5 to 6.3**

Why you cannot choose between them on the data

	χ^2/dof	AIC	BIC
M4 (mechanistic)	5.79/1	15.79	**14.75**
power law	20.10/2	15.84	15.01

$\Delta\mathrm{AIC} = 0.05$

On $n = 6$ points the two models are **statistically indistinguishable**, and they disagree six-fold about the thing we actually want to know.

This is the whole argument in one line. Model selection on the calibration window carries essentially no information about the extrapolation.

So: design the experiment that settles it

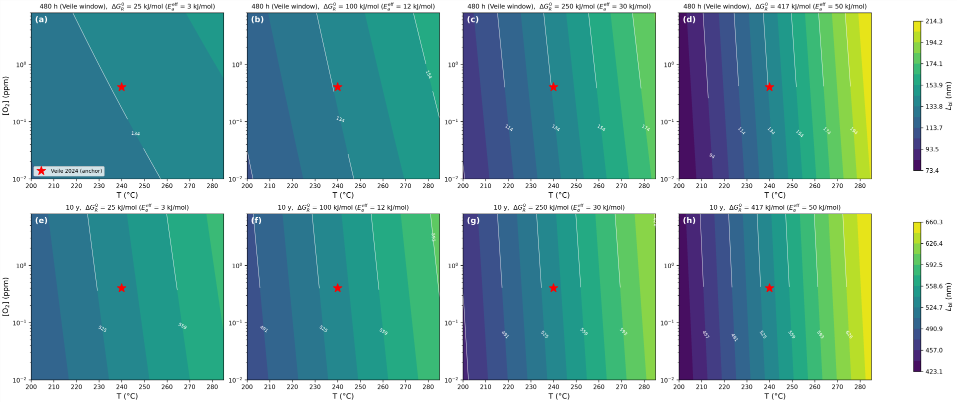
The two forms separate as the exposure lengthens. Against the measured scan-to-scan scatter:

separation criterion	exposure required
2σ, published coefficients	1437 h
3σ, published coefficients	2020 h
3σ, refit coefficients	4011 h
2σ, full envelope	$\sim$ **3000** h

Recommendation

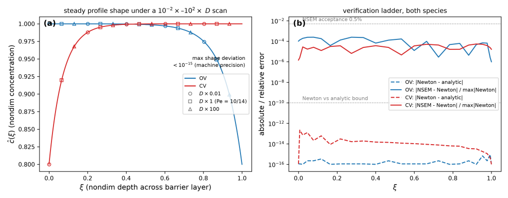
A single designed exposure of roughly 3000 h discriminates the mechanistic and empirical kinetics. That is the cheapest possible resolution, and nothing short of it will do.

Result 7: how much does the operating envelope matter?

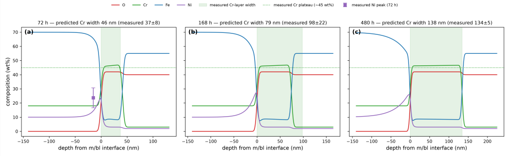


Across the scanned temperature, pressure and oxygen envelope the ten-year prediction varies by a factor of 1.19 to 1.56; small next to the 3.5 to 6.3 from the choice of model form.

Result 8: the spatial extension predicts composition with no new parameters



steady spatial solution



predicted vs measured composition

The spatial model is calibrated only by the kinetics above, then asked for depth-resolved composition and the interfacial nickel enrichment. This is a genuine out-of-sample test, and it is where the mechanistic model earns its keep over the power law.

What this study cannot claim

- **Six data points.** No amount of further computation raises the information content. We add no new experiment.
- **The fit is formally rejected.** $\chi^2 = 5.79$ on 1 dof gives $p = 0.016$; the error scale needed to make it acceptable is $s = 2.4$.
- **Part of the data is digitised** from a published figure.
- **The** $[447, 704]$ **nm interval understates the truth.** It is conditional on the model form, and the power-law comparison shows model-form uncertainty is roughly an order of magnitude larger.
- **The identifiability machinery is standard** elsewhere. The novelty is the application and the two structural findings, not the method.

Conclusions

1. A reduced point defect model fits published depth-profile data for AISI 347 in BWR hydrothermal water, but **one of five parameters is undetermined** and a second only weakly determined.
2. **95% of the fit statistic rests on three points and 80% on one.** Single-number goodness of fit conceals this.
3. The outer layer has **three parameters and three observations**. The apparent mass-balance discrepancy is a counting artefact, and the stoichiometric ratio is fully admissible.
4. Mechanistic and empirical kinetics are **indistinguishable on the calibration data** ($\Delta\mathrm{AIC} = 0.05$) yet differ **3.5 to 6.3 fold** at ten years.
5. **A designed exposure near** 3000 **h would settle it.**

The transferable point

Report identifiability alongside goodness of fit, or the extrapolation is not evidence.

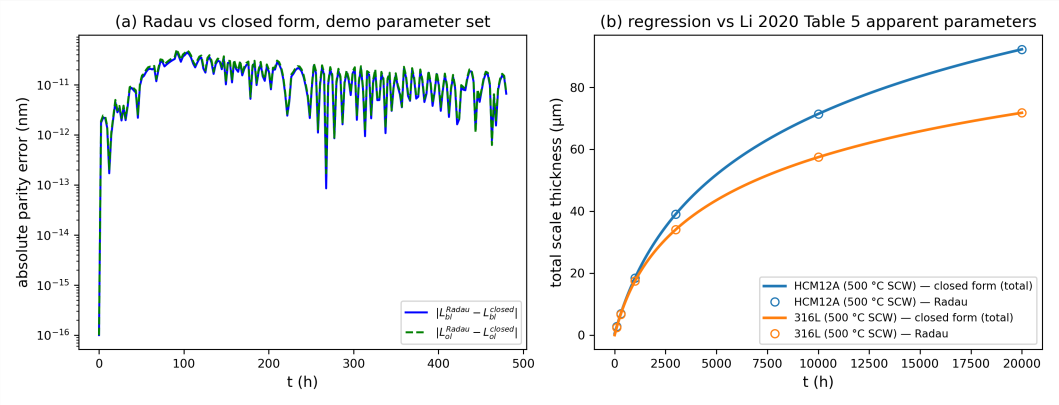
Questions

Backup: the model ladder

model	extra	n_{free}	χ^2	A_{bl}	$\mathrm{PBR}_{\mathrm{eff}}$	$L_{\mathrm{ol},0}$
M1 core	—	4	7.15	0.7516	2.433	0
M2	$+C_{\mathrm{bl}}$	5	7.15	0.7516	2.433	0
M3	$+C_{\mathrm{bl}} + C_x$	6	4.11	955.8	4.873	0
M4 (accepted)	$+L_{\mathrm{ol},0}$	5	**5.79**	0.7269	1.096	115.6
M5	$+L_{\mathrm{ol},0} + C_{\mathrm{bl}}$	6	5.79	0.7269	1.096	115.6
power law	k, n per layer	4	20.10	—	—	—

M3 reaches a lower χ^2 by running onto a degenerate ridge ($A_{\mathrm{bl}} \approx C_{\mathrm{bl}} \approx 956$, $b_3 \to 0$), which is why it is not accepted. M5 recovers M4 exactly, confirming C_{bl} is redundant.

Backup: solver verification



Numerical against closed form, agreement to better than 1×10^{-10} nm.

Backup: reproducing every number

All quantities on these slides come from committed artefacts:

slide content	artefact
M4 parameters, profiles	`outputs/paper/table1_parameters.csv`
model ladder	`outputs/paper/table2_model_ladder.csv`
bootstrap, AIC, power law	`outputs/paper/referee_stats.json`
$\mathrm{PBR}_{\mathrm{eff}}$ closure, Fe release	`outputs/paper/table10_pbr_closure.csv`

The command behind each artefact is listed in `docs/REPRODUCE.md`.